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KEMENTERIAN SUMBER MANUSIA
ARAS 7 & 8 BLOK D4, KOMPLEK D
PUSAT PENTADBIRAN KERAJAAN PERSEKUTUAN
62530 PUTRAJAYA**

KERTAS PENERANGAN (INFORMATION SHEET)

KOD DAN NAMA PROGRAM / <i>PROGRAM'S CODE AND NAME</i>	(Z-009-2:2015) CORE ABILITIES	
TAHAP / LEVEL	2 (TWO)	
NO. DAN TAJUK KEMAHIRAN / <i>CORE ABILITY NO. AND TITLE</i>	CA02 INTERPERSONAL BEHAVIOUR	
NO. DAN PENYATAAN KEMAHIRAN / <i>ABILITY NO. AND STATEMENT</i>	02.01 DEMONSTRATE RESPONSIBILITY AT WORK PLACE 02.02 APPLY TEAMWORK COOPERATION CONCEPT 02.03 INITIATE PROBLEM SOLVING SKILLS AT WORKPLACE 02.04 DEMONSTRATE WORK PERFORMANCE AWARENESS	
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TAJUK / TITLE: PROBLEM SOLVING SKILLS

TUJUAN / PURPOSE:

This Information Sheet is intended to describe on the interpersonal behavior within organization and its related regulatory affecting the usage. The person who is competent in this CA shall be able to infuse and alert of problem solving skills at workplace and aware of any regulatory acts related to it.

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PENERANGAN / INFORMATION:

1. NATURE AND STAGES OF PROBLEM SOLVING

Effective problem solving usually involves working through a number of steps or stages outlined below.

1.1 Problem Identification:

This stage involves detecting and recognising that;

- If there is a problem
- Identifying the nature of the problem and
- Defining the root cause of the problem.

The first phase of problem solving may sound obvious but often requires more thought and analysis. Identifying a problem can be a difficult task in itself,

- Is there a problem at all?
- What is the nature of the problem?
- Are there in fact numerous problems?
- How can the problem be best defined?

By spending some time defining the problem you will not only understand it more clearly yourself but should be able to communicate its nature to others. This leads to the subsequent phases.

1.2 Structuring the Problem:

This stage involves;

- Period of observation.
- Careful inspection of the matter.
- Fact-findings and.
- Developing a clear picture of the problem.

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Following on from problem identification, structuring the problem is all about gaining more information about the problem and increasing its understanding. This phase is all about fact finding and analysis, building a more comprehensive picture of both, the goal(s) and the barrier(s). This stage may not be necessary for very simple problems but is essential for problems of a more complex in nature.

1.3 Looking for Possible Solutions:

During this stage you will generate a range of possible causes of action, but with little attempt to evaluate them at this stage.

From the information gathered in the first two phases of the problem solving framework it is now time to start thinking about possible solutions to the identified problem. In a group situation this stage is often carried out as a brain-storming session, letting each person in the group express their views on possible solutions (or part solutions). In organisations different people will have different expertise in different areas and it is useful, therefore, to hear the views of each concerned party.

1.4 Making a Decision:

This stage involves careful analysis of the different possible causes of action and then selecting the best solution for implementation.

This is the most complex part of the problem solving process. Following on from the previous steps it is now time to look at each potential solution and carefully analyse it. Some solutions may not be possible, due to other problems, like time constraints or budgets. It is important at this stage to also consider what might happen if nothing is done to solve the problem - sometimes trying to solve a problem that leads to many more problems requires very creative thinking and innovative ideas.

Finally, make a decision on which cause of action to take - decision making is an important skill in itself and we recommend that you refer to pages on **decision making**.

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1.5 Implementation

This stage involves accepting and carrying out the chosen cause of action.

Implementation means acting on the chosen method of solution. During implementation more problems may arise especially if identification or structuring of the original problem is not carried out fully.

1.6 Monitoring/Seeking Feedback

The last stage is about reviewing the outcomes of problem solving over a period of time, including seeking feedback as to the success of the outcomes of the chosen solution.

The final stage of problem solving is concerned with checking that the process was successful. This can be achieved by monitoring and gaining feedback from people affected by any changes that occurred. It is good practice to keep a record of outcomes and any additional problems that occurred.

1.7 Brainstorming

Brainstorming is one of the most commonly used techniques for generating a large number of ideas in a short period of time. Whilst it can be done individually, it is more often practised in groups.

Before a brainstorming session begins, the leader or facilitator encourages everyone to contribute as many ideas as possible, no matter how irrelevant or absurd they may seem.

There should be lots of large sheets of paper, Post-IT notes and/or flip charts available, so that any ideas generated can be written down in such a way that everyone present can see them.

i. The Rules of Brainstorming

- a. Absolutely no criticism of suggestion or person is allowed. Positive feedback for all ideas should be encouraged.
- b. The aim is to produce as many ideas as possible.
- c. The aim is to generate a sense of creative momentum. There should be a feeling of excitement in the group with ideas being produced at a rapid pace.

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All ideas should be encouraged, regardless of how irrelevant, 'stupid' or 'off the mark' they might seem.

- d. Ideas should cross-fertilise each other, in other words everyone should continually look at the suggestions of the rest of the group and see if these spark any new ideas. Each person is then feeding off the ideas of the others.

2. EFFECTIVE PROBLEM-SOLVING TECHNIQUES

2.1 Outline

Every group at one time or another has the task of problem solving. Problem solving is an ongoing process that is an integral part of work and life. One of the reasons groups encounter difficulty in solving problems is that they fail to follow an organized procedure. For working groups to be effective, they should have an agreed upon procedure for problem solving. Not all problems are equally important and their relative significance should be kept in perspective. Common sense about practicality and priorities are important ingredients in maximizing people's time, energy, and commitment level to solving problems.

There are many different versions of problem solving models, but essentially they all contain the following key steps.

2.2 Problem Solving Method

i. Define The Problem

A good problem definition states the current situation and the desired situation. For example, customer satisfaction has been rated below the acceptable level or less than 5% of the targeted 20%. When defining the problem, the statement should not imply any solutions or causes.

The current situation is defined by facts, like customer satisfaction rated poor or less than 5% of historic structures. Use clear and concise language supported by gathered facts when defining the problem.

The desired situation, or what we want to happen, is defined as an objective, like 20% of eligible historic structures. The desired situation should be stated in clear, concise, concrete language and be both realistic and worthwhile.

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A primary source of group ineffectiveness is rushing to solutions before properly defining the problem.

ii. Identify And Define The Root Causes

To resolve problems so they remain solved, the group must identify and define its root causes i.e. the reasons for the problem, rather than symptoms i.e something that indicates the presence of the problem. A technique to use when considering all possible causes is brainstorming. Brainstorming is a technique that allows a large number of ideas to be generated. Prior to brainstorming, go over the ground rules with group members.

2.3 Generate Alternative Solutions

Now, that the group has identified and defined root causes and brainstorm alternative solutions. The focus of this step is to generate, not evaluate. Usually groups that generate and evaluate together often reduce the number of potentially viable solutions i.e. when one member suggests a solution, other members offer reasons why it will not work. If the group have exhausted of ideas for alternative solutions, combine aspects of previously generated solutions. Some solutions are likely to integrate the best aspects of various ideas and also are more likely to find consensus within the group.

2.4 Evaluate the Alternatives

Before evaluating alternatives, the group needs to first establish criteria for judging solutions. The criteria should define general characteristics that a solution should have. One way to help members develop criteria is to ask them to complete sentences that naturally lead to identifying criteria. The criteria should be objective and preferably measurable rather than emotional. Group members should focus only on those criteria that are needed or desired to solve the problem. Unnecessary criterion needlessly reduces the number of potential solutions. Weighing the solutions against criteria can be done with various degrees of structure. For a highly structured approach, group can develop a matrix with each of the criteria listed on one axis and each alternative solution on the other.

The criteria can be weighed if some are more important than the others. For each solution, each member assigns a score to each criterion based on the extent to which that solution meets the criterion. For example, use Linkert scale of one to ten with ten being the highest score. Ask team members to assign a value to each solution based

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on how closely it meets the criterion. Multiply the ratings for each solution by the weightage given to each criterion. Add the products for each solution and write the sum in a total score column. The solution with the highest total score is the group's choice.

2.5 Agree On The Best Solution

After completing step 4, the group is now left with the best possible solution either by evaluating criteria or voting by the unfamiliar method. Agreeing on the best solution requires that the group have some ground rules for making decisions, such as consensus or majority vote.

If the group is having difficulty in reaching consensus, members or a facilitator should work to clarify specific areas of disagreement and then identify ways to integrate similar interests into the solution. Controversies and disagreements if used constructively can be effective tools in selecting the best solution. Sometimes, as a result of disagreement, the group may revert to identify new solutions supported by group consensus.

2.6 Develop an Action Plan

Action planning is designed to involve people, build their commitment, and increase the likelihood that the solution will be implemented effectively and on time. To help structure the action plan, groups can choose from a variety of action plan types. Questions that need to be answered during action planning include;

- What they want to do (the goal)
- How each step is to be undertaken (the strategy)
- When they want to do it (timeframe)
- Who is responsible for each step and
- The expected outcome of the completed step

Typically, a group's energy and attention may decline during these steps if members think selecting the best solution is the peak of the problem-solving process. As long as the groups understand, up front, that action planning is part of the process of problem solving. Members of the groups should accept this phase as part of the process.

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2.7 Implement and Evaluate the Solution

Solution steps should be implemented according to the action plan. However, sometimes unintended consequences occur that require changing the implementation plan midstream. The group can adapt to unintended consequences by suggesting that project updates or status be added to each meeting's agenda so that the group has a regular way to check its progress.

Evaluation is probably the step most groups under emphasize when it comes to problem solving. The reasons why groups do not conduct evaluations are time, cost, political climate, uneasiness in evaluation, and fear of being challenged in their decision. However, some evaluation is essential in determining the effectiveness of the solution. An evaluation can take many forms and be as simple as brief conversations with key individuals or can be elaborated as sophisticated surveys and in-depth record analysis.

The purpose of evaluation is to determine how well the solution is working or why the solution may not be working. The group should decide how comprehensive the evaluation should be based upon the complexity of the problem and the decided solution. Depending on the severity of the problem and the quality of solution desired, objectives and approaches for problem solving will vary. For minor problems, a group may focus on quickly reaching a solution as the main objective. Groups concerned with finding a quality solution as the main objective may cause in more time for the problem solving process.

To be effective, problem solving must arrive at a solution that gets the job done, efficiently uses the available resources, promotes cooperation, and fosters competence among those faced with solving the problem. Problem solving is an ongoing process that is an integral part of everyday life either at home or at work. A problem must first be felt, understood, and alternative choices created before effective groups can consider its solution, implement the decision, and evaluate how well the solution is working.

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Questions

- 1 Briefly state on how to come up with problem solving?

(5 marks)

- 2 State the stages for Problem Structuring.

(4 marks)

- 3 How do you decide in solving a problem?

(2 marks)

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